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FLEXURE DESIGN AND TESTING FOR STC STIRLING CONVERTORS

Songgang Qiu*, Allen A. Peterson†, and Jack E. Augenblick†
Stirling Technology Company, Kennewick WA

Abstract. Stirling Technology Company (STC) is a leading developer of high efficiency, high reliability Stirling convertors for both space and commercial applications. STC has been working with the Department of Energy and NASA Glenn Research Center (GRC) to evolve this technology for many years. One key to the success of STC's robust dynamic machines is their unique use of flexure bearings. STC flexures are optimized for each system application. STC relies on flexures to provide the majority of the axial spring rate, to provide radial spring rate to resist piston rubbing, and to provide mechanical centering forces without the use of centering gas ports. This paper provides some general design criteria that STC uses when designing a new flexure. Axial stiffness, radial stiffness, subassembly rocking modes, and flexure arm modes must all be considered for each new flexure design. This paper discusses the impact of critical flexure parameters to flexure behavior and system performance, and compares flexure test data to analytical predictions.

INTRODUCTION

Advanced free-piston Stirling convertors, cryocoolers, and linear motor/compressors are the core business of STC. The capability for combining highly efficient energy conversion with maintenance-free and degradation-free operation makes the STC Stirling machines and linear motors attractive for a myriad of applications. These include remote power generation for rural villages, remote cabins, and transmission towers. Its silent operation is suitable for recreation vehicles, residential combined heating and power generation, embassy backup power in under-developed countries, etc.[6] The free-piston Stirling convertor has long been considered a candidate for a variety of space power applications in part because of the demonstrated long life and high efficiency[3]. There has been increasing interest in developing a radioisotope free-piston Stirling convertor for deep space exploration, with reduced isotope inventory (plutonium), due to its high efficiency.

The key technology that is critical to long life and maintenance-free operation, and that differentiates STC free-piston Stirling generators from others is the unique implementation of flexure bearings both for Stirling displacer assemblies and linear alternators. The proven benefits of STC free-piston, flexure-bearing Stirling machines include: 1) the elimination of rubbing seals and lubrication, 2) elimination of gas-bearing port plugging failures, 3) good predictability, repeatability, reliability and stability, and 4) minimized performance and operating frequency variations due to variability in operating conditions. STC convertors have been shown to be efficient, reliable, robust, and capable of safe operation over various thermal input at nearly constant system efficiency and wide environment conditions, from extremely cold weather in Alaska to boiling hot temperature in Arizona desert[1,2,3,4,5]. An STC's RemoteGen™ 10-W convertor (RG-10) has been running over 76,000 hours with no maintenance and no degradation. More than 200 RG-10 convertor and displacer flexures, at 65% over-stroke, have operated without failure for over 80,000 hours.

Flexure bearings for Stirling engines have been used successfully for more than 30 years. The planar flexures are made from sheet stock, using a variety of methods to cut spiral or other types of kerfs that form one or more support legs between clamped regions near the outside and inside diameters of the flexure, as shown in Figure 1. In a typical linear alternator flexure bearing application, the outside of the flexure bearing is clamped to the stationary stator assembly, and the inside portion is clamped to the mover (plunger) rod (moving ID flexure bearing). In a typical STC displacer flexure bearing configuration, the inside portion of the flexure is clamped to a stationary displacer rod/pole, and the outside of the flexure is connected to the moving displacer piston (moving OD flexure bearing).

Normally two bearings (or flexure stacks) are attached to each moving member with axial separation to provide a support-couple wheelbase (moment arm), preventing the moving member from rocking, as shown in Figure 2. STC's spiral flexures enable the moving member to spring axially with relative freedom, but the wide flexure arms provide a very high resistance to radial motion. The flexure and moving member are a mass-spring dynamic system. The system frequency is largely dependent upon the total spring rate of the flexure stacks. The spiral flexure's extremely high radial spring rate enables the moving member to

* Director of Engineering
† Advanced Mechanical Engineer

operate with a tight clearance seal of 12 to 25 microns, depending on specific configuration.



Figure 1. Spiral Flexure and Clamping

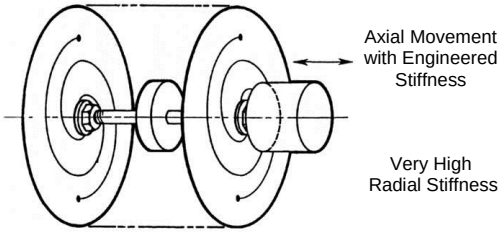


Figure 2. Typical Flexure Bearing

Flexures are designed for infinite life, which is achievable by maintaining the maximum stress on the flexure, below an allowable endurance limit. The allowable stress for a selected material is obtained by subtracting 3σ (three times standard deviation) from vendor's mean endurance limit, resulting in a 99% confidence level.

Flexure reliability is further improved by inspecting every flexure with a digital microscope and burn-in testing every flexure on a flexure test rig for at least of 10^7 cycles. For burn-in test, flexures are operated at 20 to 30% over-stroke for additional safety margin. After buildup, each Stirling convertor has to go through a minimum of 100 hours of infant mortality test ($>2.16x10^7$ cycles) to account for assembly bias and to verify the convertor system integrity.

FLEXURE DESIGN ANALYSIS

There have been many different types of flexure bearings reported in literature. The type of linear motor used in the Stirling largely affects the correct flexure choice. Spiral arm designs are the most commonly-used flexure bearings. Spiral flexures usually provide very high and circumferentially uniform radial spring rate, which is especially suitable for high side-load, no torsion-load STC moving iron linear motors. Alternatively, linear-arm flexure designs usually have higher axial spring rate than spiral flexures, but the radial spring rate uniformity is usually poor. Spiral flexure bearings may not be suitable for other linear alternator designs, which have low side-load but inherently high torsion-load. These designs often use straight, cantilever flexures or linear suspension elements.

This paper presents only the design analysis and testing data for STC spiral flexures. The design approaches and some special design considerations discussed in this paper are, however, applicable to most other styles of flexure design.

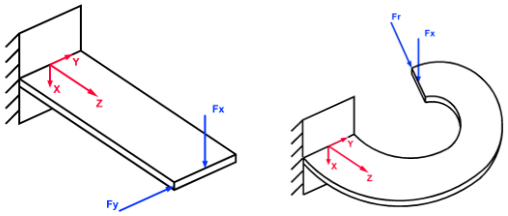
Before personal computers (PC) and Finite Element Analysis (FEA) software were widely available, flexures were analyzed as a cantilevered beam as shown in Figure 3a. Dimensional analysis was often used to derive dimensionless equations for the flexure. These dimensionless equations and parameters, along with testing and FEA analysis, helped to understand and express the fundamental behavior of flexures of any type, as discussed later. With rapid growing of the PC and wide availability of FEA software, flexure arm shapes, stresses, and spring rates are currently optimized with the FEA optimization utility.

The spiral flexure arms are simulated with a curved cantilevered beam, as shown in Figure 3b. The closed form solutions are difficult to achieve, but the computer modeling can often predict the spring rates within 10% of the tested values. For a curved beam, an axial load on the free-end results in not only a bending load as shown, but also a torsion load, which leads to higher stress at the edges of the fixed-end.

The mathematical model for a spiral flexure can be expressed as Equation 1.

$$R = R_i + (R_o - R_i)*[\theta/\theta_0 + f*\sin(2\pi\theta/\theta_0)] \quad (1)$$

Where R is the polar distance from the center of the flexure, θ/θ_0 is the relative sweep angle ratio, R_i is the active inner radius, R_o is the active outer radius, and f is the arm shape factor described later in Figure 7. Figure 4 shows these characteristics of a flexure graphically.



a)Straight Beam b)Curved Beam

Figure 3. Cantilevered Beams

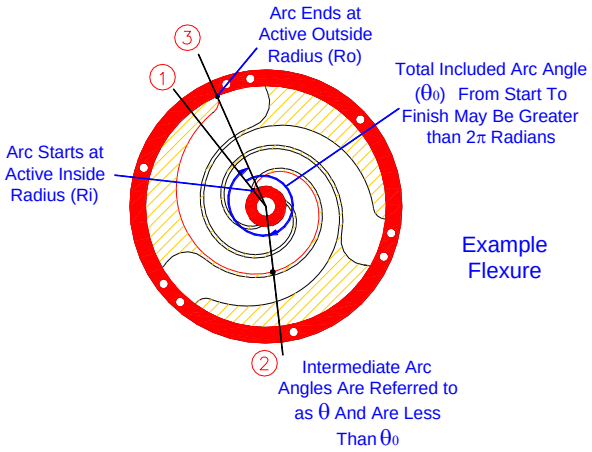
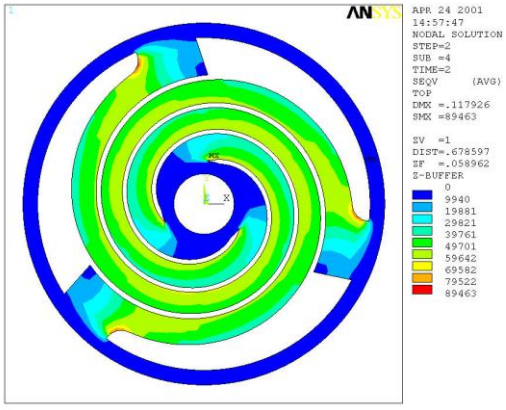
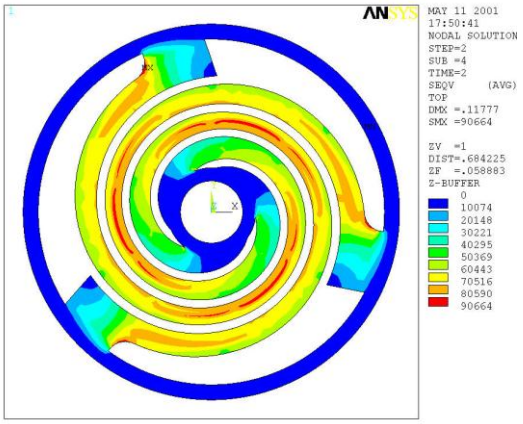


Figure 4. An Example Flexure

For a typical Stirling engine design, the flexure active inner and outer radii, and amplitudes are determined from the Stirling optimization and linear alternator design trades. General approach is to minimize the size and weight. The required axial spring rate is predicted with linear dynamics analysis and the required radial spring rate is obtained to address side load and rocking mode. Trade studies are often conducted to determine the best combination of size, thickness, amplitude, and the number of flexure arms. FEA structural analysis code has been used at STC for flexure optimization for over 14 years. Figure 5 shows the stress distributions for (a) a non-optimized flexure and (b) an optimized flexure. Non-optimized flexures often have highly concentrated stresses at both ends of the flexure arms, whereas optimized flexures tend to distribute the stress over the entire length of the arm. Flexure thickness plays an important role in achieving evenly-distributed stress.



a) Highly-Concentrated Stress

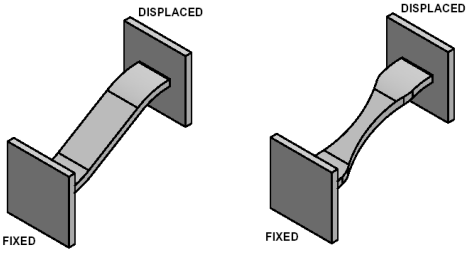


b) Well-Distributed Stresses

Figure 5. Flexure Stress Distributions

Arm Shape Factor "f"

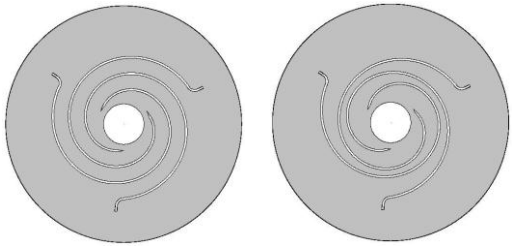
The large stress concentration at both ends of the flexure arms is similar to what happens if one end of a straight beam is clamped and displaced a distance while the other end of the beam is fixed, as in Figure 6a. More evenly distributed stress over the beam can be achieved by gradually varying the beam width as shown in Figure 6b.



a) Straight Beam b)Varying Beam Width

Figure 6. Varying Beam Width

For a spiral flexure, varying arm width is realized by a non-zero arm shape factor, f, in Equation 1, as shown in Figure 7. In STC flexure design code, the shape factor, f, is one of the optimization parameters, which are adjusted until the best configuration is achieved.

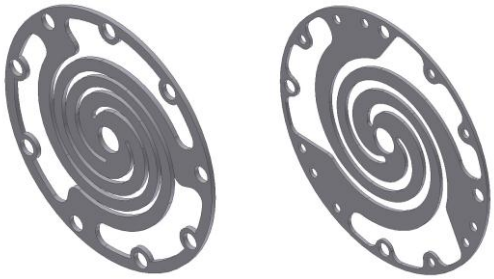


a) f=0 b) f=0.1

Figure 7. Flexures with Different f Factors

Flexure Kerf Widths

In designing STC flexures, kerf width (gap between arms) is often adjusted. Larger kerf widths usually yield lower spring rates as well as less moving mass. With appropriate design, net system mass and vibration are reduced. Figure 8 shows two flexures, one with a narrow kerf and the other with a wide kerf.



a) Narrow Kerf b) Wide Kerf

Figure 8. Flexures with Different Kerfs

A spiral flexure with multiple arms represents a multi-degree-of-freedom system with many modes. Modal analysis can be used to calculate modal frequencies. The objective of the modal analysis is to make sure that none of the modal frequencies are near the operating frequency, or near the multiples of the operating frequency. Increasing kerf often results in higher modal frequencies. Table 1 lists the calculated modal frequencies of two flexures with kerf width 0.9 mm (0.0354 in.) and 1.7 mm (0.0682 in.), respectively. Both flexures are designed at 1.6 mm (0.062 in.) thick and 14 mm (0.551 in.) amplitude.

Table 1. Calculated Modal Frequencies

Kerf width	0.0354in	0.0682in
1st Mode	82 Hz	90 Hz
2nd Mode	133	144
3rd Mode	133	144
4th Mode	193	247
5th Mode	193	247

Modal Frequencies

The 1st mode is the flexure natural frequency, when it is supporting no additional mass. This frequency has to be greater than the operating frequency so that each flexure can reciprocate some additional mass. Higher flexure natural frequency indicates a smaller number of flexures will be required to achieve the designed operating frequency. The 2nd and 3rd modes represent the flexure rocking frequency about the flexure in-plane axes, as shown in Figure 9. The 4th and 5th modes are the most important ones, which represent the frequency of irregular flexure arm motion, as depicted in Figure 10. The frequency of this irregular arm motion is often called "surge" frequency. When surge frequency corresponds to one of the flexure harmonics, the arms of neighboring flexures can contact each other, resulting in quick failure.

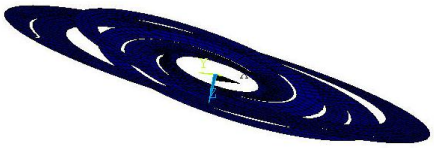


Figure 9. Flexure Rocking Mode

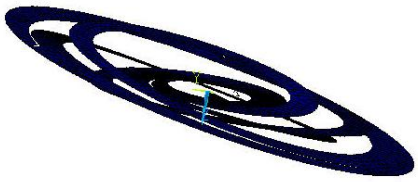


Figure 10. Flexure Surge Mode

There are many other modes that could also cause flexure arm contact, fretting, and beating or chiseling noises. The general rule of STC flexure design is to make sure that there is absolutely no arm contact at any modal shape. At STC, besides the FEA modal analysis being performed, visualization methods are commonly used to survey the flexure arm motion. STC large amplitude flexures are often optimized with non-linear (large deflection) analysis, but the modal shapes are always calculated with linear analysis, as only linear modal analysis utility is available in any FEA software. For a visual survey testing, a stack of flexures are attached to a flexure test rig and swept from a few Hertz below to a few Hertz above the designed

frequency. The arm motions through the survey frequency band are carefully recorded. Table 2 is an example of the survey summary for a 3-arm flexure designed for 14 mm amplitude and 60 Hz operation. Spaces marked "OK" indicate no irregular motions.

Table 2. Visual Survey of the Arms of a Flexure

Xp (mm)	Frequency (Hz)							
	56	57	58	59	60	61	62	63
12.7	OK	SL	SS	SS	SL	MS	LS	ML
10.2	OK	SL	SS	SL	SL	LS	LS	ML
7.62	SL	SS	SL	OK	SL	LS	ML	SL

Letter Codes of Motion	
First Letter	Second Letter
S small	S (Surge, multiple arm movements)
M medium	L (Lag, arms continue to move at top and bottom)
L large (arm contact)	Q (Quiver, high frequency arm movement)

From Table 2, it can be concluded that this specific flexure can only be used at an operating frequency of 60 Hz or lower.

Multiple Arm Design

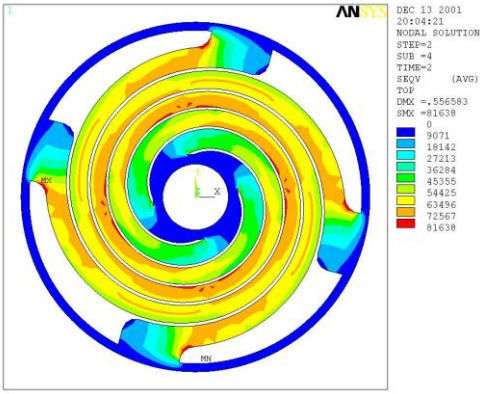
Three-arm spiral flexures are the most commonly used. Two-arm and four- or more arm flexures are also seen and can be helpful, dependent on design requirements. Adding arms to a flexure usually results in a higher axial spring rate, but lower radial stiffness. Table 3 shows the comparison of a three-arm flexure to four-, six-, and eight-arm flexures of the same physical size, amplitude, and thickness. Examples of four-arm and eight-arm flexures are shown in Figure 11.

Dimensional analysis based on plate and beam results in the relationships shown as Equation 2.

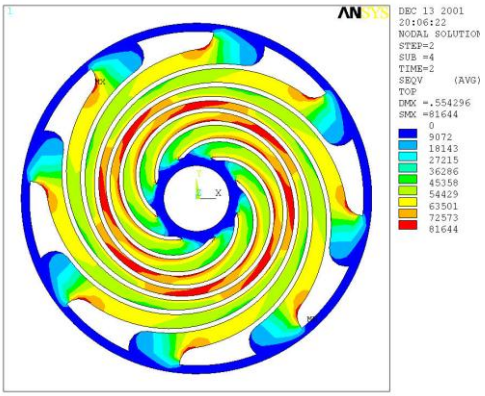
$$K_{axial} \sim t^3/D^2 \qquad \text{and} \qquad K_{radial} \sim t \qquad (2)$$

Where K_{axial} is the axial spring rate, K_{radial} is the radial spring rate, t is the flexure thickness, and D is the active OD.

Flexure testing at STC indicates that flexures behave much differently from what is in Equation 2. For example, K_{axial} is proportional to the flexure diameter, instead of inversely proportional to the square of the diameter as indicated in Equation 2. Tests have also shown that the flexure diameter significantly affects the radial spring rate.



a) A four-arm flexure



b) An eight-arm flexure

Figure 11. Multiple-Arm Flexures

Table 3. Comparison of multiple-arm flexure spring rates

	3 arms	4 arms	6 arms	8 arms
K_{axial} (N/m)	25643	31138	37000	43960
K_{radia} (N/m)	838964	758970	651146	583931

Calculations have shown that increasing flexure arms from three to four usually results in higher axial spring rate. Further increase of the flexure arm may or may not help the axial spring rate, dependent upon the material thickness and flexure diameter. Generally, for a small diameter flexure of 2 inches or below, maximum axial spring rate occurs at four to five arms design. For big diameter flexures of 6 inches or above, the axial spring rate usually peak out at eight or more arms.

ADDITIONAL DESIGN CONSIDERATIONS

In designing flexures, considerations are also given to other aspects of the Stirling system, such as displacer and power piston/mover rod rocking modes. If the

rocking mode frequency is near the operating frequency or near the multiples of the operating frequency, displacer or piston rubbing would occur. NASA GRC performed a hammer impact testing of an RG-55 Stirling convertor in May 2001[5]. A modal response of displacer around 134 Hz was clearly identified in the test. A displacer rocking-mode is schematically explained in Figure 12. This rocking frequency and associated damping are deemed appropriate, as an STC RG-55 successfully passed vibration test at launch load of 0.2 g^2/Hz (12.3 g rms) at NASA GRC in 1999.

As part of continuous technology advancement efforts for improving reliability, mass reduction, and vibration reduction, STC's RG-55 displacer moving mass was reduced compared to the original design. This required a reduction in axial spring rate and removal of some displacer flexures from original build, to maintain the same design frequency. The rocking frequency will be sustained if the flexure radial spring rate is increased through flexure redesign (with thinner sheets) but with the same total flexure stack height, or with a reduced arm-count configuration. The same rocking mode can also be achieved if the wheelbase is enlarged via a longer spacer between the two stacks of flexures with less radial spring rate.

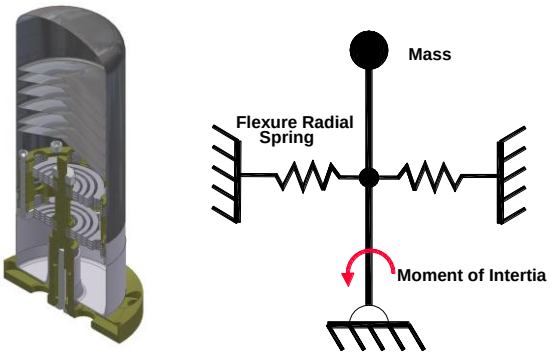


Figure 12. Displacer and Its Rocking Mode

The piston/mover assembly exhibits the same rocking mode as for the displacer, but at a much higher frequency. A lighter piston is always better, as the cantilevered piston contributes the most to the moment of inertia (Figure 2). No rocking mode was identified for the RG-55 convertor piston/mover motion in GRC impact testing[5]. Calculations showed that this rocking mode frequency is high and of no concern for the STC RG-55 convertor. A good design practice is always to check for rocking mode. With further advancement in weight reduction and fewer flexures required for reduced moving mass, rocking mode should be addressed appropriately. For some other linear alternators, flexures provide only a few percent of the total required spring forces, and most of the spring forces are from gas springs. In that case, rocking mode could be problematic.

OVERALL DESIGN OPTIMIZATION

The STC flexure design approach produces reliable, high-performance flexures that are optimized for each overall system application. STC flexures are designed to consider all the issues discussed in this paper together. STC carefully designs its flexures for uniform material usage, to produce flexures with very high axial spring rates without compromising available radial spring rate.

CONCLUSIONS

STC spiral flexures are fully optimized with FEA optimization utilities. The STC design methodology considers all aspects of system implementation, rather than optimizing for only one parameter.

Flexure design is often iterated with Stirling modeling and linear alternator design analysis for best performance and lowest weight/cost. Good design of the flexure results in more evenly distributed stress over the entire arms, instead of highly concentrated stress at the ends of the arms.

Good use of shape factor can help achieve evenly distributed stress over the entire arm. A thicker material usually results in higher effective axial spring rate, but lower effective radial spring rate.

More flexure arms could mean higher axial spring rate, dependent on the flexure thickness and diameter. A lower radial spring rate always occurs for more arms in similarly-configured flexures.

Modal frequencies of a flexure have to be designed away from the system operating frequency, or harmonics of the system frequency, to avoid flexure arm contact. The kerf width can be used to reasonably adjust the flexure modal frequency. Larger kerf width usually means higher modal frequencies.

Displacer and piston/mover motion rocking modes need to be evaluated to avoid potential displacer or piston rubbing due to rocking. This rocking mode frequency has to be different from the system operating frequency or the multiples of the operating frequency.

ACKNOWLEDGEMENTS

Special thanks are due the STC engineering staff for its continuous dedication to the advancement of STC flexure design and analysis. The success of STC Stirling convertors is the direct result of the implementation of state-of-art STC flexures, which evolved through 30 years of innovations. STC also appreciates NASA GRC for their long-lasting interest in Stirling technology and continuous support technically and financially.

REFERENCES

1. Qiu, Songgang., Peterson Allen A., "Developing a Free-Piston Stirling Power Convertor for Advanced Radioisotope Space Power Systems," in *Proceedings of STAIF-2002 (Space Technology and Applications International Forum)*, Albuquerque NM, *3*-6 February 2002.
2. Qiu, Songgang, Peterson, Allen A., "Stirling Convertor Performance Mapping Test Results," in *Proceedings of STAIF-2002*, Albuquerque NM, *3*-6 February 2002.
3. Schreiber, Jeffery G., " Assessment of The Stirling Power Option for Space Science Application," *Proceedings of 2000 Aerospace Conference*, IEEE, Big Sky MT, March 2000.
4. Thieme, L.G. and Schreiber, J.G., " NASA GRC Stirling Technology Development Overview," in *Proceedings of STAIF-2003*, Albuquerque NM, February 2003.
5. Vicente J. Suarez, Thomas W. Goodnight, William O. Hughes, Sergey Samorezov, "Impact Testing of a Stirling Convertor Linear Alternator," in *Proceedings of STAIF-2002*, Albuquerque NM, *3*-6 February 2002.
6. White, Maurice A, Colenbrander, Kelvin, Olan, Ronald W., and Penswick, L. Barry, "Generators That Won't Wear Out," ASME *Mechanical Engineering*, Vol. 118, No. 2, February 1996.

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